

Steps

3. $PC(Ann)$

4. $CS(Ralph) \rightarrow PC(Ralph)$

5. $\neg CS(Ralph)$

Reason

Premise

Universal instantiation on 1

Modus tollens on 2,4

Valid conclusion: Ralph is not a CS major.

e) "What is good for corporations is good for the United States."
 "What is good for the United States is good for you."
 "What is good for corporations is for you to buy lots of stuff."

Soln: This is an ambiguous question.

f) "All rodents gnaw their food." "Mice are rodents." "Rabbits do not gnaw their food." "Bats are not rodents"

Soln: $Ro(x)$: x is a rodent $Gn(x)$: x gnaws their food.
 $Mi(x)$: x is a mice $Ra(x)$: x is a rabbit $Ba(x)$: x is a Bat

Premise: ① $\forall x(Ro(x) \Rightarrow Gn(x))$ ② $\forall x(Mi(x) \Rightarrow Ro(x))$
 ③ $\forall x(Ra(x) \Rightarrow \neg Gn(x))$ ④ $\forall x(Ba(x) \Rightarrow \neg Ro(x))$

Reason

Premise

Premise

Premise

Premise

Universal instantiation on 1

Universal instantiation on 2

Hypothetical syllogism on 6,5

Universal generalization on 7

Steps

1. $\forall x(Ro(x) \Rightarrow Gn(x))$

2. $\forall x(Mi(x) \Rightarrow Ro(x))$

3. $\forall x(Ra(x) \Rightarrow \neg Gn(x))$

4. $\forall x(Ba(x) \Rightarrow \neg Ro(x))$

5. $Ro(c) \Rightarrow Gn(c)$

6. $Mi(c) \Rightarrow Ro(c)$

7. $Mi(c) \Rightarrow Gn(c)$

8. $\forall x(Mi(x) \Rightarrow Gn(x))$